

विद्युत ऊर्जा / Electrical Energy

Total installed Power Capacity (31st January 2026):- 5,20,511 MW, comprising of 2,48,542 MW (47.7%) of fossil-fuel sources and 2,71,969 MW (52.3%) of non-fossil fuel sources. Share of 'Renewable Energy' sources in total installed capacity is 2,63,189 MW (50.6%).

Contribution of various energy sources (in percentage):-

Fossil Fuel

Coal based thermal power:- (2,21,210 MW) 42.5% ; Gas :- 3.9%; Diesel :- 0.1%; Lignite Coal (6,620 MW):- 1.3%

Non-Fossil Fuel (Renewable Energy Sources)

Total solar energy :- 27.0 % ; Large Hydro Power:- 9.8%

Total Wind Energy :- 10.5% ; Bio Power (जैव शक्ति) :- 2.1%

Nuclear Energy :- 1.7% ; Small Hydropower Project:- 1.00%

Top 5 states in terms of installed capacity of coal based thermal power generation

Uttar Pradesh > Maharashtra > Chhattisgarh > Madhya Pradesh > Gujarat

Top states in terms of total renewable energy installed capacity

Rajasthan > Gujarat > Tamil Nadu > Karnataka > Maharashtra

Top states in terms of installed hydroelectric power capacity

Himachal Pradesh > Uttarakhand > Karnataka > Jammu & Kashmir > Maharashtra

Top States in Terms of Installed Nuclear Power Capacity:

Tamil Nadu > Gujarat > Maharashtra > Rajasthan > Karnataka > Uttar Pradesh

परमाणु ऊर्जा संस्थापित क्षमता (विश्व):- USA > China > France > Russia (India - 12th)

Note: France meets 70% of its total energy requirements through nuclear energy which is maximum in the world.

Top states in terms of solar power installed capacity:-

Rajasthan > Gujarat > Maharashtra > Tamil Nadu > Karnataka

Solar Energy Installed Capacity (Global) :- China > USA > India > Japan > Germany

Ground mounted Solar Energy (जमीन पर स्थापित सौर्य ऊर्जा उत्पादन में शीर्ष राज्य)

Rajasthan > Gujarat > Tamil Nadu > Karnataka > Maharashtra

Top States in Rooftop Solar Energy (PM-Surya Ghar Yojana).

Gujarat > Maharashtra > Rajasthan > Kerala

Top states in terms of wind energy installed capacity :- Gujarat > Tamil Nadu > Karnataka > Maharashtra > Rajasthan

Wind Power Installed Capacity (Global) :- China > USA > Germany > India

Iron -Ore

Iron ore reserves (India) :- Karnataka > Odisha > Jharkhand > Chhattisgarh

Iron ore production (India):- Odisha > Chhattisgarh > Karnataka > Jharkhand

Iron ore reserves (world):- Australia > Brazil > Russia > China > India

Iron Ore Production (World):- Australia > Brazil > India > China

Iron ore reserve states and their regions

State	Major iron ore regions
Odisha	Gurumahisani, Kiruburu, Badam Pahar (Keonjhar and Mayurbhanj districts)
Jharkhand	Noamundi, Badaburu, Pansiraburu (West Singhbhum)
Chhattisgarh	Dalli Rajhara, Bailadila (Durg and Dantewada districts), Bilaspur
Madhya Pradesh	Balaghat, Chhindwara, Jabalpur, Jhabua
Maharashtra	Nagpur, Bhandara, Ratnagiri
Karnataka	Chikmagalur (Baba Budan Hill), Kudremukh
Goa	Adulmale, Kudnam
Andhra Pradesh	Ongole, Guntur

Bauxite

Odisha ranks first in terms of bauxite reserves and production in India.

Top Producers in India:- Odisha > Gujarat > Jharkhand > Chhattisgarh

The largest reserves of bauxite in the world:- Guinea > Australia > Brazil

The largest producers of bauxite in the world:- Australia > Guinea > China > Brazil

Aluminium Production (World) :- China > India > Russia > Canada

Major bauxite producing/Reserve areas in India

State	Region/Districts
Odisha	Kalahandi, Sambalpur, Koraput, Bolangir.
Gujarat	Gulf of Kutch (Jamnagar), Gulf of Khambhat (Kheda)
Maharashtra	Kolhapur, Thane, Ratnagiri
Jharkhand	Ranchi, Palamu, Lohardaga
Chhattisgarh	Bilaspur, Durg
Tamil Nadu	Salem

Copper

copper reserves :- Rajasthan > Jharkhand > Madhya Pradesh

copper production :- Madhya Pradesh > Rajasthan > Jharkhand

Major copper producing areas

Madhya Pradesh: Malajkhand (Balaghat)

Jharkhand: Ghatshila, Mosabani (East Singhbhum)

Rajasthan: Khetri (Jhunjhunu)

Andhra Pradesh: Guntur, Nellore, Kurnool

Chile ranks first in the world in both copper reserves and production.

Silver

Leading states in terms of silver reserves:- Rajasthan > Jharkhand > Andhra Pradesh > Karnataka

silver production :- Rajasthan > Karnataka

Silver reserves (world):- Peru > Australia > China > Poland

Silver Production (World):- Mexico > China > Peru > Poland

Gold

According to the National Mineral Data Year book, the total estimated reserves of gold ore (primary) in the country are 501.83 million tonnes. The **largest reserves are in Bihar (44%)**, Rajasthan (25%), Karnataka (21%), West Bengal (3%), Andhra Pradesh (3%), Jharkhand (2%) respectively.

Karnataka state alone produces 93% of the total gold produced in the country. **Kolar and Hatti gold mines located in Raichur district of Karnataka are famous for gold production.**

Gold reserves (world):- Australia > Russia > South Africa > USA

Gold Production (World):- China > Russia > Australia > Canada

The world's largest gold mine:- Nevada Gold Mine (USA), Grasberg Gold Mine (Indonesia), Olimpiada Gold Mine (Russia).

Gold reserves in Bihar:- Karmatia Village, Sono Block, Jamui Subdivision, District-Jamui.

****** India's first large - scale Pvt. Gold Mining Venture since Independence - Jonnagiri Gold Mine, Kurnool (Andhra Pradesh).**

Estimated reserve:- 50 Tonnes of Gold.

Pvt. Company- Geomysore Services India Pvt. Ltd.

Commercial production by this pvt. Company is presently underway- A proud moment for India's Gold Sector.

Mica

Andhra Pradesh produces the maximum amount of mica (41%). India ranks first in the world in the production of mica. The mica belt in India is concentrated in Bihar, Jharkhand, Rajasthan and Andhra Pradesh. **India's top mica belt is spread in Hazaribagh, Munger, Gaya. Koderma, located in this belt, is called the capital of mica.** USA is the largest importer of mica produced by India.

mica producing areas

Bihar:- Munger, Gaya, Bhagalpur, Jamui, Nawada

Jharkhand:- Hazaribagh, Koderma, Singhbhum, Palamu

Rajasthan:- Jaipur, Udaipur, Bhilwara, Ajmer, Kishangarh

Andhra Pradesh:- Nellore, Khammam, Visakhapatnam

Diamond

Diamond is mainly found in igneous rocks and alluvial deposits. At present, the diamond areas of India are divided into four regions:-

- 1. South India:-** Anantapur, Kadapa, Guntur, Krishna, Mahbubnagar and Kurnool districts in the state of Andhra Pradesh.
- 2.** Behradin, Kodavalli area in Raipur district and Bastar district of Chhattisgarh.
- 3.** Panna district in Madhya Pradesh.
- 4.** Between the Mahanadi-Godavari valleys in the state of Odisha.

Production (India):- Madhya Pradesh > Andhra Pradesh > Chhattisgarh
90.18%

Diamond reserves (world):- Russia > Botswana > Congo > South Africa

Diamond Production (World):- Russia > Botswana > Canada > Congo > South Africa

Uranium

Uranium Ore:- Uranium dioxide or uraninite or pitchblende.

Uranium ore fields in the country

Jharkhand: Jadugoda, Bhatin, Narwapahar, Turmadih, Bagjata.

Uttar Pradesh: Lalitpur

Meghalaya: Domiasat

Rajasthan: Rohail

Karnataka: Gogi

Andhra Pradesh: The **Tumalapalli mine in Kadapa district is the largest uranium mine in the country.**

Uranium reserves (world):- Australia > Kazakhstan > Canada > Russia
28% 13% 10% 8%

(The world's largest single-location uranium deposit – Olympic Dam, South Australia)

Uranium Production (World) :- Kazakhstan > Canada > Namibia > Australia > Uzbekistan
39% 22%

Coal

Types of coal:-

Anthracite:- Carbon content up to 90-95%

Bituminous:- Carbon content up to 55-80%

Lignite:- Only 40-55% of carbon

peat :- Carbon content less than 40%

Coal reserves (India):- Jharkhand(26.4%) > Odisha(25.1%) > Chhattisgarh(18.1%) > West Bengal(10.8%) > Madhya Pradesh(9.8%) > Telangana(7.3%) > Maharashtra(4.2%) > Bihar(1.4%)

Coal production (India):- Odisha > Chhattisgarh > Jharkhand > Madhya Pradesh > Telangana

Coal reserves (world):- USA > Russia > Australia > China > India

Coal Production (World):- China > India > Indonesia > USA > Australia

Petroleum/Crude Oil

crude oil production in India

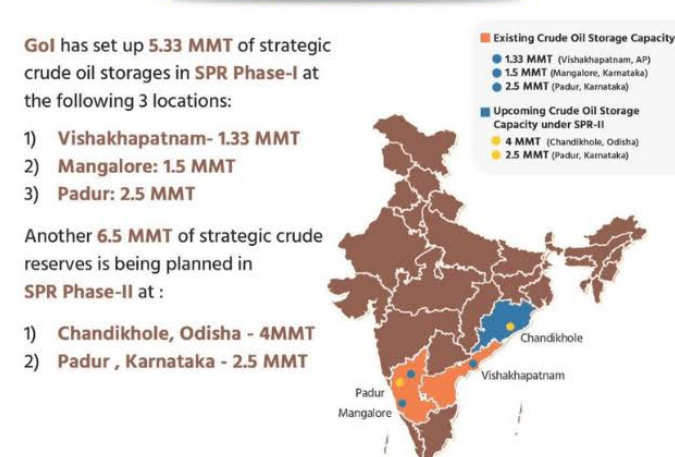
Offshore Area (Mumbai High) > Rajasthan > Gujarat > Assam > Tamil Nadu

Natural Gas

Top natural gas producing regions/states in India

Offshore areas (Mumbai High and Basin) > Assam > Rajasthan > Tripura > Gujarat

India strategic Petroleum Reserves Ltd(ISPRL)



Monazite:- It is a mixture of thorium, uranium, cerium and lanthanum etc. The third stage of India's nuclear power programme is thorium based. India has the world's largest monazite deposits which have high amount of thorium. It is found in large quantity in the coastal sand of Palakkad and Kollam districts of Kerala.

By :- **Aseem Srivastava**